

Naive Baye's Algorithm (Classification)

① Probability

② Baye's Theorem

Independent Events

Rolling a Dice $\{1, 2, 3, 4, 5, 6\}$

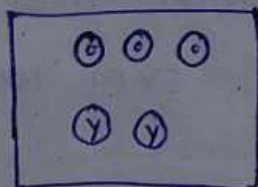
$$P_0(1) = \frac{1}{6}$$

$$P_0(2) = \frac{1}{6}$$

$$P_0(3) = \frac{1}{6}$$

Dependent Events

① What is the probability of removing a orange marble and then a yellow marble.

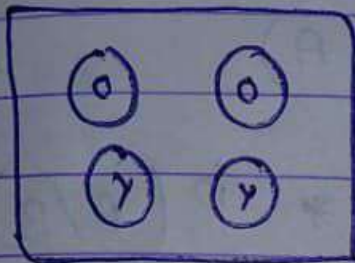


$$\rightarrow P(O) = \frac{3}{5} \rightarrow 1^{st} \text{ Event} \rightarrow$$

↓

orange marble has been Removed.

$$P(Y/O)$$



$$\rightarrow P(Y) = \frac{2}{4} = \frac{1}{2} \Rightarrow 2^{\text{nd}} \text{ Event}$$

↓
Removed the yellow marble.

$$P(O \text{ and } Y) = P(O) * P(Y/O)$$

↑
1st

↑
2nd

Conditional Probability

$$= \frac{3}{5} * \frac{1}{2} = \boxed{\frac{3}{10}}$$

$$P(A \text{ and } B) = P(A) * P(B/A)$$

Baye's Theorem

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) * P(B/A) = P(B) * P(A/B)$$

$$P(A/B) = \frac{P(A) * P(B/A)}{P(B)}$$

↓
Baye's Theorem

$P(A/B)$ = Probability of Event A
given B has occurred.

$P(A)$ = Probability of Event A

$P(B)$ = Probability of Event B

E/P feature

x_1 x_2 x_3			Dependent
-	-	-	Yes
-	-	-	No
-	-	-	Yes
-	-	-	No

$$P(Y/(x_1, x_2, x_3)) = \frac{P(Y) * P(x_1, x_2, x_3)/Y}{P(x_1, x_2, x_3)}$$

$$= \frac{P(Y) * P(x_1/Y) * P(x_2/Y) * P(x_3/Y)}{P(x_1) * P(x_2) * P(x_3)}$$

$$[0.60, 0.40]$$

↓
Yes

New Test Data

$$P_{\star}(Yes/(x_1, x_2, x_3)) = \frac{P(Yes) * P(x_1/Yes) * P(x_2/Yes) * P(x_3/Yes)}{P(x_1) * P(x_2) * P(x_3)}$$

↳ Constant

$$= [0.60]$$

↓
Yes

$$P_{\star}(No/(x_1, x_2, x_3)) = \frac{P(No) * P(x_1/No) * P(x_2/No) * P(x_3/No)}{P(x_1) * P(x_2) * P(x_3)}$$

↳ constant

$$= [0.40]$$

↓
No

Note: Remove or cut Denominator because it is same for both Yes or No.

lets solve this Problem

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	weak	No
2	Sunny	Hot	High	strong	No
3	Overcast	Hot	High	weak	Yes
4	Rain	Mild	High	weak	Yes
5	Rain	Cool	Normal	weak	Yes
6	Rain	Cool	Normal	strong	No
7	Overcast	Cool	Normal	strong	Yes
8	Sunny	Mild	High	weak	No
9	Sunny	Cool	Normal	weak	Yes
10	Sunny	Mild	Normal	weak	Yes
11	Rain	Mild	Normal	strong	Yes
12	Sunny	Mild	High	strong	Yes
13	Overcast	Hot	Normal	weak	Yes
14	Rain	Mild	High	strong	No

Outlook

	Yes	No	$P(E/Yes)$	$P(E/No)$
Sunny	2	3	$2/9$	$3/5$
Overcast	4	0	$4/9$	$0/5$
Rain	3	2	$3/9$	$2/5$

Temperature

78

	Yes	No	$P(E/Yes)$	$P(E/No)$
Hot	2	2	$2/9$	$2/5$
Mild	4	2	$4/9$	$2/5$
Cool	3	1	$3/9$	$1/5$

Play (Y/N)

	Yes	No	$P(Yes)$	$P(No)$
	9	5	$9/14$	$5/14$

→ Test (Sunny, Hot) → O/P

$$P(Yes / (Sunny, Hot)) = \frac{P(Yes) * P_2(Sunny/Yes) * P_2(Hot/Yes)}{P_2(Sunny) * P_2(Hot)}$$

$$= \frac{9/14 * \frac{2}{9} * \frac{2}{9}}{7/14 * \frac{2}{9}}$$

$$= \frac{2}{63} = \underline{\underline{0.031}}$$

$$P(\text{No} / (\text{Sunny, Hot})) = \frac{P(\text{No}) * P(\text{Sunny} / \text{No}) * P(\text{Hot} / \text{No})}{P(\text{Sunny}) * P(\text{Hot})}$$

$$= \frac{1}{14} * \frac{3}{7} * \frac{2}{5}$$

$$= \frac{3}{35} = \underline{\underline{0.085}}$$

$$P(\text{Yes} / (\text{Sunny, Hot})) = \frac{0.031}{(0.031 + 0.085)}$$

$$= 0.27 = 27\%$$

$$P(\text{No} / (\text{Sunny, Hot})) = \frac{0.085}{(0.031 + 0.085)}$$

$$= 0.73 = 73\%$$

New Test 21

78

Outlook

Temperature

C/P

Sunny

Hot

73% $\Rightarrow$ They will
not play
Tennis

27% $\Rightarrow$ They will
play Tennis

Conclusion

21

0 $\Rightarrow$ Person is not going to
play Tennis